La Scuola d'Italia / Dr. Huson / IB Math 11
4 November 2026 Markscheme

2.9 Classwork: Every Rate of Change Has a Name — Markscheme

Lesson notes with worked answers; not scored, so no mark codes. Teacher notes in italics. All figures are simplified models.

1. **Sales tax** slope = tax rate (*aliquota*), constant of proportionality

   (a) $1.08875x = 174.20$, so $x = \dfrac{174.20}{1.08875} = 160$ dollars.

   Undoing the tax is the inverse function, $x = P/1.08875$. Common error: $174.20 \times (1 - 0.08875) = 158.74$, *because* 8.875% *of the total is not* 8.875% *of the price before tax.*

2. **Income tax** slope = marginal tax rate (*aliquota marginale*); I-intercept = standard deduction

   (a) $T = 0.12(40\,000 - 16\,000) = 0.12 \times 24\,000 = 2880$ dollars.

   (b) $I = 40\,000$: $\dfrac{2880}{40\,000} = 0.072 = 7.2\%$.

   $I = 100\,000$: $T = 0.12 \times 84\,000 = 10\,080$, so $\dfrac{10\,080}{100\,000} = 10.08\%$.

   The average rate is $\dfrac{0.12(I - 16\,000)}{I} = 0.12 - \dfrac{1920}{I}$. It rises toward 12% but never reaches it, and its graph is a curve, not a line.

   This is the first curve of the packet: a rational function with horizontal asymptote 0.12 *(preview of 2.8). Marginal rate ≠ average rate is a common misunderstanding about taxes.*

   (c) Tax credit: $T = 0.12(I - 16\,000) - 500$. The graph moves **down** 500 (vertical translation). Larger deduction: $T = 0.12(I - 20\,000)$. The graph moves **right** 4000 (horizontal translation). The slope, 0.12, does not change in either case.

   Link to 2.11 transformations: $f(x) - k$ shifts down, $f(x - h)$ shifts right. Italian: a deduzione reduces taxable income; a detrazione reduces the tax itself.

3. **Budget line** slope = −relative price (*prezzo relativo*), opportunity cost

   (a) x-intercept: $3x = 24$, $x = 8$: all the money on espressos (8 espressos, no cannoli). y-intercept: $4y = 24$, $y = 6$: all the money on cannoli (6 cannoli, no espressos).

   (b) $3x + 6y = 24$; slope $= -\dfrac{3}{6} = -\dfrac{1}{2}$. Now each extra espresso costs only half a cannolo, because cannoli are more expensive.

4. **Break-even** slope of C = marginal cost (*costo marginale*) = 6; slope of profit = contribution margin (*margine di contribuzione*) $= 18 - 6 = 12$

   (a) $18q = 9000 + 6q$, so $12q = 9000$ and $q = 750$ pizzas a month. $C = R = 18 \times 750 = 13\,500$ dollars.

(b) Profit $= 18(1000) - (9000 + 6 \times 1000) = 3000$ dollars.

Equivalently, profit $= 12q - 9000$*: each pizza past the 750th adds \$12.*

5. **Tariff** demand slope = price sensitivity (*sensibilità al prezzo*); share of the tariff paid by buyers = pass-through (*traslazione dell'imposta*)

(a) $2q + p = 40$ and $q - p = -16$. Adding: $3q = 24$, so $q = 8$ (thousand kg per week) and $p = 40 - 16 = 24$ dollars per kg.

(b) New supply $p = 19 + q$, so $q - p = -19$. Adding: $3q = 21$, so $q = 7$ and $p = 40 - 14 = 26$ dollars per kg.

(c) Buyers pay $26 - 24 = \$2$ of the \$3 tariff: a pass-through of $\frac{2}{3}$. Sellers keep $26 - 3 = \$23$ per kg, \$1 less than before.

The tariff is a vertical translation of the supply line (2.11). The steeper (less price-sensitive) the demand line, the more of the tariff the buyers pay. Elasticity is the same idea in percent terms; we do not compute it here.

6. **Two ways to pay tax** slopes: effective rate (*aliquota effettiva*) 0.15, marginal rate 0.25

   (a) $T - 0.15R = 0$ and $T - 0.25R = -5000$. Subtracting: $0.10R = 5000$, so $R = 50\,000$ dollars, where both taxes are $T = 7500$.

   $R = 80\,000$: flat rate $0.15 \times 80\,000 = 12\,000$; standard $0.25 \times 60\,000 = 15\,000$. The flat rate is cheaper.

   Below \$50 000 *the standard option is cheaper; above it the flat rate is cheaper, because its line is less steep. Ask the class what Italian families say about the regime forfettario.*

7. **Stocks** slope = beta (*coefficiente beta*); intercept = alpha (*alfa*)

   (a) Bank: $0.5 + 1.5(-4) = -5.5\%$. Con Edison: $0.2 + 0.5(-4) = -1.8\%$. Inverse fund: $-(-4) = +4\%$.

   (b) $x + y = 10\,000$ and $1.5x + 0.5y = 12\,000$. $0.5\times$ first: $0.5x + 0.5y = 5000$. Subtracting: $x = 7000$, so $y = 3000$. That is \$7000 in the bank and \$3000 in Con Edison.

   High beta: the stock swings more than the market (banks). Low beta: it swings less (utilities). The portfolio beta is the weighted average, $\frac{1.5(7000)+0.5(3000)}{10\,000} = 1.2$.

8. **Bonds** slope of price vs yield is negative; duration (*durata finanziaria*) is that slope in percent terms

   (a) $P \approx 100 - 7(4.5 - 4) = 96.50$ dollars.

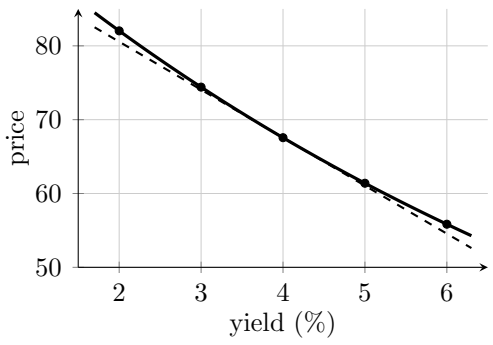
   (b) $x + y = 100\,000$ and $2x + 7y = 500\,000$. $2\times$ first: $2x + 2y = 200\,000$. Subtracting: $5y = 300\,000$, so $y = 60\,000$ and $x = 40\,000$. That is \$40 000 in the short bond and \$60 000 in the long bond.

   Same structure as the 2-7CW two-fund problem and Example 7(b): a total, plus a weighted total.

9. **Convexity** (*convessità*)

   (a)

Yield (%)	P	L
2	82.03	80.55
3	74.41	74.05
4	67.56	67.56
5	61.39	61.06
6	55.84	54.56



   (b) The curve is **above** the line on both sides of 4%. It gets **less steep** as the yield rises: from 2% to 3% the price falls 7.62; from 5% to 6% it falls only 5.55.

The curve bends up (convex, concavità verso l'alto): the slope (duration) changes, and convexity names that change. It favours the bondholder: when yields fall the price rises more than the line predicts, and when yields rise it falls less.

10. **Learning curve** learning rate 20%; economies of scale (*economie di scala*)

(a)

Pack	1	2	4	8	16	32
Cost ($)	10 000	8000	6400	5120	4096	3276.80

The first pack below $4500 is pack 16.

Each doubling multiplies the cost by 0.8*: a geometric sequence (Unit 1). The cost per pack falls quickly at first and then slowly, a curve that bends up. The total cost keeps rising but bends down, because each extra pack is cheaper.*

11. **S-curve** (*curva logistica*)

(a) First differences: 3, 7, 15, 23, 23, 15, 7, 3.

The product spreads fastest between years 3 and 5 (23 points a year). The inflection point (*punto di flesso*) is at $t = 4$, 50%.

Second differences: 4, 8, 8, 0, −8, −8, −4. *Positive (bending up) before* $t = 4$*, negative (bending down) after. The rate of change of the rate of change has a sign, and it changes at the inflection point.*